Explainable ai, attempt 13-Dec-2023 19:55

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Question 1

3D is known to be problematic for visualizing multi-dimensional numerical data because of

Correct[ ] a. Animated transitions.
Correct[ ] b. Potentially hidden information.
probably no, because its not hidden in a way that you cant find it at all....its just occluded and can be seen through rotation and zooming. IDK what the correct answer is though.
Correct[X] c. Perspective distortion effects.
Correct[X] d. The necessity to rotate the visualization for inspecting the data.
Correct[ ] e. Grouping principles.
Correct[ ] f. Change blindness.
Change blindness is problematic for animations
Correct[ ] g. Inattentional blindness.
Correct[ ] h. The user's stereo vision.

Question 2

A cluster of items (point cloud) in a 2D projection can mean that

Correct[ ] a. some attributes of the high-dimensional dataset are identical to each other
Correct[ ] b. some attributes of the high-dimensional dataset are dissimilar to each other
Correct[X] c. some attributes of the high-dimensional dataset are similar to each other
Correct[ ] d. the items are disjunct from each other

Question 3

A k-means clustering result shown in a scatterplot can be visualized by

Correct[ ] a. Connecting all cluster centroids with each other
Correct[X] b. Coloring the items by cluster
Correct[ ] c. Showing a dendrogram on top of the scatterplot
Correct[X] d. Connecting items to the cluster centroids they belong to
Correct[ ] e. Modifying the position of items according ot their cluster assignment
Correct[X] f. Assigning different shapes to items by cluster

Question 4

A scrollytelling interface shown on a public newspaper website that contains predefined slides with interactive content is an example of

Correct[X] a. Interactive slideshow
Correct[ ] b. Martini glass structure
Correct[X] c. Vodka glass structure
Correct[ ] d. Drill-down story

Question 5

A storytelling article that goes through predefined steps and then allows the user to explore the data is known as a

Correct[ ] a. Interactive slideshow
Split into multiple scenes, allow interaction mid-way
Correct[ ] b. Scrollytelling interface
Correct[ ] c. Drill-down story
No, because drill-down does not go through predefined steps. We let reader decide which path to follow, all paths are annotated.
Correct[X] d. Martini glass structure
Starts author driven (= predefined steps), then opens up for exploration

Question 6

A storytelling article that has multiple predefined substories and where the reader can freely decide which one to follow is known as

Correct[X] a. Drill-down story
Let the reader decide which path to follow, all paths are annotated. In 2
Correct[ ] b. Whiskey glass structure
Correct[ ] c. Scrollytelling interface
Correct[ ] d. Martini glass structure
Start with author driven, open up for exploration. In 2
Correct[ ] e. Interactive slideshow
Split into multiple scenes, allow interaction mid-way. In 2

Question 7

Consider an image for which your neural network predicts a class C. You use GradCAM on the image to get an explanation for class C. You notice that only the background region of the image is highlighted in the GradCAM explanation. Which of the following conclusions can you safely draw from the explanation?

Correct[ ] a. The background region in the image is similar to that of other images of class C.
Probably no
Correct[ ] b. The background region is generally important to classify an image as class C.
Probably yes
Correct[ ] c. Changes in the background region are most likely to affect the prediction of class C for most images
Correct[X] d. None of the three conclusions can be safely drawn.

Question 8

In Permutation Feature Importance

Correct[ ] a. we compute how much of a model's output variance is explained by the features.
Error of model is used, not variance. In 5
Correct[X] b. random feature permutation leads to the destruction of information.
In 5
Correct[X] c. performance on altered datasets is measured.
In 5
Correct[ ] d. features are permuted to lie in a normalized range for comparison.
In 5
Correct[ ] e. low relative error means the feature is important.
High relative error means the feature is important. In 5

Question 9

On which model classes can we apply influence functions?

Correct[X] a. Linear Regression
Correct[ ] b. k-Nearest Neighbor
Correct[X] c. Neural Network
Correct[ ] d. Decision Tree

Question 10

The following are disadvantages of dimensionality reduction techniques:

Correct[X] a. It can be difficult to preserve the semantics of single attributes/dimensions
Correct[X] b. The results can be hard to interpret.
Correct[ ] c. Outliers in the data cannot be taken into account.
Correct[X] d. The calculation can be time-consuming.
yes, there is visual analytics exam 22.2.21 where not ticking this resulted in loss of points. (t-SNE is slow)
Correct[X] e. Visual patterns in the lower dimensional embedding can be misleading.
Correct[ ] f. Categorical attributes cannot be taken into account.

Question 11

The following are examples where machine learning is used for visualization (ML4Vis):

Correct[ ] a. a visualization that shows potentially interesting patterns identified by a pre-trained models
Correct[X] b. a machine learning model that is used to detect usage patterns of a visualization tool
Correct[X] c. a machine learning model that is used to recommend next analysis steps in a visualization tool
Correct[X] d. a machine learning model that is used to to suggest suitable port types for a given dataset
Correct[ ] e. a network that represents the graph structure of a neural network
Correct[ ] f. a visualization that shows the latent space of a deep neural network
Correct[ ] g. a heatmap that highlights pixels of an input image that most caused the output classification
Correct[X] h. a machine learning model that is used to detect recurring patterns in time-series data

Question 12

The following are properties of the t-SNE dimensionality reduction

Correct[ ] a. shapes in the data are preserved in the embedding space
Correct[X] b. cluster sizes in the embedding space can be meaningless
Correct[X] c. distances between clusters in the embedding space can be meaningless
Correct[ ] d. random noise in the data looks random in the embedding space

Question 13

The following are purposes of explainable AI

Correct[X] a. Teaching & understanding concepts
Correct[ ] b. Training models
Correct[X] c. Interpretability & explainability
Correct[ ] d. Applying AutoML approaches
Correct[ ] e. Generating models
Correct[X] f. Comparing and selecting models

Question 14

The following are valid approaches for handling large datasets

Correct[X] a. Aggregate the data before visualzing it
Aggregation & Filtering
Correct[X] b. Combine visualization with algorithmic/automated techniques
Correct[X] c. Allow users to interact with the data
Part of temporal partitioning: pan, rotate, zoom
Correct[X] d. Sample the data before visualzing it
See https://www.computerweekly.com/de/definition/Data-Sampling-Datenauswahl
Correct[X] e. Spatially partition the data before visualizing it
Spatial partitioning: Multiple coordinates views Overview + detail
Correct[X] f. Temporally partition the data before visualizing it
Temporal partitioning: Navigation: pan, rotate, zoom Geometric vs. semantic zooming

Question 15

The following are valid statements for ConfusionFlow and InstanceFlow techniques

Correct[ ] a. ConfusionFlow operates on the instance level.
No, on class level
Correct[X] b. ConfusionFlow shows the per-class confusion as heatmaps.
Correct[X] c. InstanceFlow shows the evolution of items using a Sankey diagram.
Correct[X] d. ConfusionFlow operates on the class level.
Correct[X] e. InstanceFlow is able to show the classification of single instances over multiple epochs.
Correct[ ] f. InstanceFlow is able to visualize the overall performance of the network during training.
Correct[ ] g. IntanceFlow is able to show which parts of images contribute to the classification result.
No, GradCam does that.
Correct[ ] h. InstanceFlow is able to rank single instances by various scores/measures.
Correct[ ] i. ConfusionFlow does not provide any global performance measures.
Correct[ ] j. InstanceFlow operates on the global level.
No, on the instance level.
Correct[X] k. InstanceFlow visualizes instances using the LineUp technique.
See https://www.youtube.com/watch?v=EAyvPtf84ns
Correct[X] l. InstanceFlow shows the evolution of instance class confusion using a Sankey diagram.
Correct[ ] m. ConfusionFlow shows the evolution of class confusion using a stream graph.
No, just line charts and heatmaps.
Correct[ ] n. ConfusionFlow shows the evolution of class confusion using a Sankey diagram.
No, just line charts and heatmaps.

Question 16

The following are valid statements for patterns that may appear in the Time Curves visualization

Correct[X] a. Patterns can be misleading because of the projection method applied
Correct[X] b. Very similar versions of the same Wikipedia article appear as a cluster pattern
In 3
Correct[X] c. Back and forth editing between two versions of the same Wikipedia article appears as an alternation pattern
In 3
Correct[ ] d. Big edits/changes of the same Wikipedia article appear as a cluster pattern
They appear as an oscillation. In 3
Correct[X] e. Big edits/changes of the same Wikipedia article appear as an oscillation pattern
In 3

Question 17

The following are valid statements for the Time Curves technique:

Correct[X] a. spatial proximity in the visualization indicates similarity between time points
in caption of Fig 1 in https://hal.inria.fr/hal-01205821/document they write Fig. 1. The time curve principle: a) a timeline is folded into itself in such a way that similar time points end up being close to each other; b) Example: a time curve showing the evolution of a Wikipedia article. In my opinion not really clearly formulated. They should have written: Time points with similar context end up close to each other. Streit formulated the answer similar to the figure caption. So I will tick it in the exam and complain if they mark it as false.
Correct[X] b. the temporal ordering of the data is preserved
Correct[ ] c. the layout is independent of the projection method used
Correct[X] d. can be applied to visualize textual changes in documents

Question 18

The following are valid statements in the context of deep learning visualization

Correct[X] a. heatmaps are commonly used for showing instance-specific features
Correct[ ] b. heatmaps are commonly used for showing the architecture of a network
Correct[ ] c. stream graphs are commonly used for showing the architecture of a network
https://en.wikipedia.org/wiki/Streamgraph
Correct[X] d. heatmaps are commonly used for showing attribution
Attribution maps (aka. saliency maps) are subtype of heatmaps. Example: GradCAM.An attribution map signals pixels in the input image highly relevant to the downstream tasks or hidden representations, and therefore highlights the "attention" of a network over a specific task. In other words, the model space can be thought as a space defined on top of attribution maps, where the affinity between points or networks is evaluated using the distance between their produced attribution maps, which again, requires no supervision and can be computed really fast.
Correct[ ] e. heatmaps are commonly used for showing spatial attribution
Usually things are cut out
Correct[X] f. heatmaps are commonly used for showing confusion between classes
Yes, in confusion matrices and ConfusionFlow

Question 19

The following statements are valid for Anscombe's quartet

Correct[ ] a. Consists of five datasets with X and Y tuples
Correct[ ] b. Consits of four datasets with three dimensions
Correct[X] c. Used as a motivating example for the value of visualization
Correct[X] d. The datasets share important statistical properties
Correct[X] e. Consits of four datasets with X and Y tuples
Correct[X] f. The datasets look very different when being plotted
Correct[ ] g. Consits of five datasets with two dimensions each

Question 20

The following statements are valid for particle-based visualizations

Correct[ ] a. Are regularly used for visualizing Anscombe's quartet
Tricky again. Watch lecture 2, ~1h9min. There he is saying particle based visualization can be compared to snow falling down or sedimentation. In Anscombe's quartet we have a (x, y) tuple. However, the lecture notes say: idea: Show every data point as a single particle. Asking Marc Streit he answered that answer a is incorrect, although not saying if my reasoning is correct.
Correct[ ] b. Represent every data attribute/dimension as a single particle
No, every data item is a single particle.
Correct[X] c. Are based on sedimentation effects that can be observed in nature
In 2
Correct[X] d. Represent every data item as a single particle
In 2
Correct[X] e. Are regularly used for visualizing Bayes' theorem
In 2, in video at ~1h9min
Correct[ ] f. Are regularly used for showing outliers in time-series data
Correct[X] g. Can be used to show progressive changes in streaming data
Probably yes

Question 21

The following statements are valid for the Matchmaker visualization technique

Correct[X] a. It draws connection bands between heatmaps
https://www.youtube.com/watch?v=vi-G3LqHFZA
Correct[X] b. It relies on the continuation grouping principles
https://www.youtube.com/watch?v=vi-G3LqHFZA
Correct[ ] c. It represents tabular data using a scatterplot matrix
https://www.youtube.com/watch?v=vi-G3LqHFZA
Correct[ ] d. It represents tabular data using a radar chart
https://www.youtube.com/watch?v=vi-G3LqHFZA
Correct[X] e. It allows users to compare clustering results
https://www.youtube.com/watch?v=vi-G3LqHFZA
Correct[X] f. It represents tabular data using parallel coordinates
https://www.youtube.com/watch?v=vi-G3LqHFZA

Question 22

The following statements are valid for the data-ink ratio by Edward Tufte:

Correct[ ] a. should be minimized for producing effective visualizations
Correct[ ] b. should be O to produce effective visualizations
Should be 1
Correct[X] c. should be understood as a general guideline rather than a strict optimization criterion
Correct[X] d. chart junk decreases the ratio
Correct[ ] e. chart junk increases the ratio
Correct[X] f. should be maximized for producing effective visualizations

Question 23

The following statements are valid in the context of data-driven storytelling

Correct[ ] a. A drill-down story starts author-driven and then opens up for exploration
No, it does not start author-driven. We let reader decide which path to follow, all paths are annotated.
Correct[ ] b. An interactive slideshow starts author-driven and then opens up for exploration
Author driven -> open up is a Martini glass structure
Correct[X] c. A purely reader-driven story does not order the parts of the story in a specific way
Correct[X] d. A purely author-driven story does not allow the reader to interact with the information
For example, a drill-down story is purely reader-driven.
Correct[ ] e. A purely reader-driven story orders the parts of the story in a specific way
Correct[ ] f. A purely reader-driven story does not allow the reader to interact with the information

Question 24

The following statements are valid in the context of dimensionality reduction techniques

Correct[ ] a. t-SNE and PCA are linear approaches
PCA and MDS are linear. t-SNE is non-linear.
Correct[X] b. t-SNE requires setting hyperparameters
Correct[X] c. UMAP is computationally faster than t-SNE
Correct[ ] d. PCA is computationally slower than t-SNE
No, it's faster.
Correct[X] e. MDS is a linear approach
PCA and MDS are linear. t-SNE is non-linear.

Question 25

The following statements are valid in the context of grouping principles

Correct[X] a. grouping principles can overrule each other
connection is a very strong grouping principle and can overrule others
Correct[ ] b. proximity and reliability are both grouping principles
Not reliability
Correct[ ] c. continuation is a grouping principle used in scatterplots
Correct[X] d. connection is a grouping principle used in node-link representations
Correct[X] e. bump charts rely on the continuation grouping principle
Correct[X] f. node-link representations rely on the connection grouping principle
Correct[X] g. clusters of items in scatterplots rely on the proximity grouping principle
Correct[X] h. bump charts rely on the connection grouping principle
Correct[ ] i. the continuation grouping principle overrules similarity and proximity
Correct[ ] j. scatterplots rely on the continuation grouping principle

Question 26

The following statements are valid in the context of visualizing time-series (time-dependent) data

Correct[X] a. Animation can make it hard to compare data from different time points
Correct[X] b. The progression of the data over time can be shown using animation
Correct[X] c. The data at different time points can be shown next to each other
Verified by Marc Streit: Yes, that's correct. Juxtaposing the visualizations for the different time points can actually be very effective in many scenarios because it allows easy comparison (without the need to interpret an animation).
Correct[X] d. The time aspect can be encoded using color of other visual channels

Question 27

The following visualization techniques can typically be used to represent the functioning of sorting algorithms

Correct[ ] a. attribution maps
Attribution maps are subtype of heatmaps. Example: GradCAM.An attribution map signals pixels in the input image highly relevant to the downstream tasks or hidden representations, and therefore highlights the "attention" of a network over a specific task. In other words, the model space can be thought as a space defined on top of attribution maps, where the affinity between points or networks is evaluated using the distance between their produced attribution maps, which again, requires no supervision and can be computed really fast.
Correct[X] b. heatmaps
Correct[ ] c. the ConfusionFlow technique
Confusion Flow visualizes NN Confusion across epochs
Correct[X] d. bump charts
Classical quicksort, etc. can be explained via bump charts. In 2
Correct[X] e. node-link diagrams
Classical topological sort explanation is via node-link diagrams
Correct[ ] f. the Matchmaker technique
The Caleydo Matchmaker technique allows users to split multidimensional datasets into subgroups, cluster them separately and analyze relations between the resulting clusters

Question 28

What are the differences between "Concept Bottleneck Models" and "TCAV" (Testing with Concept Activation Vectors)?

Correct[ ] a. "TCAV" is intrinsically interpretable.
TCAV is posthoc, therefore not intrinsically interpretable
Correct[X] b. "TCAV" measures the relationship between a concept and a class.
Correct[X] c. "Concept Bottleneck Models" produce human understandable concepts, which are then used to make a prediction.
Correct[X] d. "Concept Bottleneck Models" are intrinsically interpretable.

Question 29

What are the differences between "Counterfactual Explanations" and "Adversarial Examples"?

Correct[X] a. "Counterfactual Examples" are intuitive explanations for humans
Correct[ ] b. "Adversarial Examples" are used to interpret a model
In 5
Correct[X] c. "Adversarial Examples" aim to deceive a model
In 5
Correct[ ] d. "Counterfactual Examples" optimize a loss function to generate examples, but "Adversarial Examples" do not optimize a loss function
No, because adversarial examples also optimize a loss function. (REF: https://christophm.github.io/interpretable-ml-book/adversarial.html)

Question 30

What is true about Influential Instances?

Correct[ ] a. Cannot give local explanations.
Correct[ ] b. Cannot give global explanations.
Correct[X] c. Can give global explanations.
Correct[X] d. Can give local explanations.

Question 31

Which XAI Methods give global explanations?

Correct[X] a. Prototype and Criticism
Correct[X] b. Activation Maximization
Correct[ ] c. Counterfactual Explanations
Correct[ ] d. Nearest Neighbor Examples

Question 32

Which XAI methods are usually considered to give local explanations?

Correct[X] a. Saliency maps
Aka. attribution maps, sensitivity maps. A subtype of heatmaps. Plots the most important features in an image capture. GradCAM.
Correct[X] b. Concept Bottleneck Models (CBM)
Correct[ ] c. Testing with Concept Activation Vectors (TCAV)
TCAV gives global explanations
Correct[X] d. LIME

Question 33

Which of the following statements about LIME are true?

Correct[ ] a. LIME is equivalent to a global surrogate model for small values of the locality parameter.
I would say opposite.
Correct[X] b. The surrogate models used for LIME have to be one-dimensional.
According to Marc Streit, this is true.
Correct[X] c. LIME uses local, weighted models fit on randomly sampled data.
Perturbing the dataset involves random sampling
Correct[ ] d. LIME is robust against adversarial attacks.
No, opposite

Question 34

Which of the following statements about Network Dissection are true?

Correct[X] a. Network Dissection is based on comparing regions of strong activation to concept-labeled image segments.
Correct[X] b. Network Dissection is a model-specific explanation technique.
Correct[ ] c. Network Dissection assigns concepts to units.
No, units are matched to concepts. Network Dissection is an interpretability method for CNNs that evaluates the alignment between individual hidden units and a set of visual semantic concepts. By identifying the best alignments, units are given human interpretable labels across a range of objects, parts, scenes, textures, materials, and colors.
Correct[ ] d. Network Dissection yields global insights about the network based on activation statistics.
No, can be local or global depending on usage. Network Dissection is an interpretability method for CNNs that evaluates the alignment between individual hidden units and a set of visual semantic concepts. By identifying the best alignments, units are given human interpretable labels across a range of objects, parts, scenes, textures, materials, and colors.
Correct[X] e. Network Dissection matches units to concepts.
Correct[X] f. Network Dissection can yield local insights about the network based on activation statistics.
Can yield global or local depending on usage
Correct[ ] g. Network Dissection assigns units to concepts.
It matches, not assigns

Question 35

Who are the potential users of deep learning visualizations according to the classificaiton by Hohman et al.?

Correct[X] a. Model users
Correct[ ] b. Agents
Correct[ ] c. Data engineers
Correct[X] d. Model developers & builders
Correct[X] e. Non-experts
Correct[ ] f. Domain experts